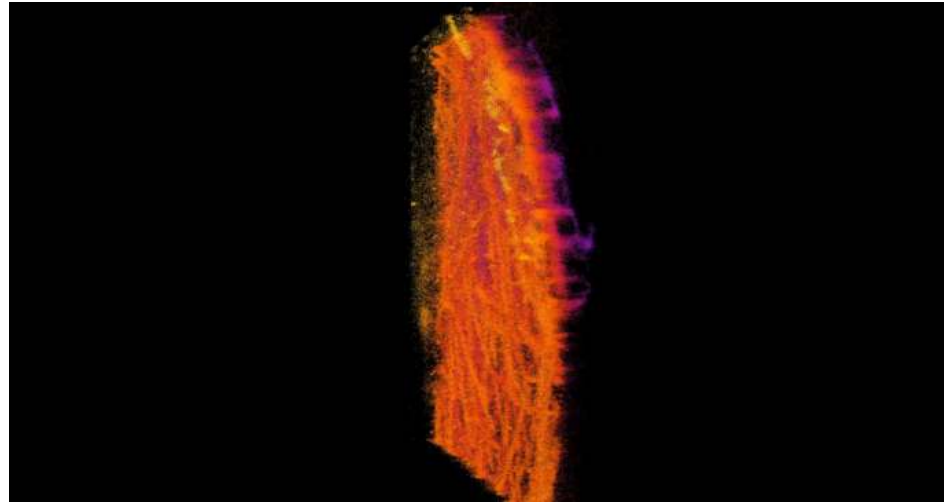


Learning-methods for single molecule localization microscopy

Romain Séailles



PDF export — this is a static version of an interactive presentation; some slides may appear incomplete or degraded.

Light microscopy

Light microscopy is a technique that uses **visible light** and a system of lenses to generate magnified images of small objects.



1 mm



100μm



10μm



1μm



100nm



Optical microscope

Light microscopy

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Abbe diffraction limit (1873)

The minimum resolvable distance is

$$d = \frac{\lambda}{2n \sin \theta} = \frac{\lambda}{2 \text{NA}}.$$

For a conventional setup, $d \simeq 200\text{nm}$.



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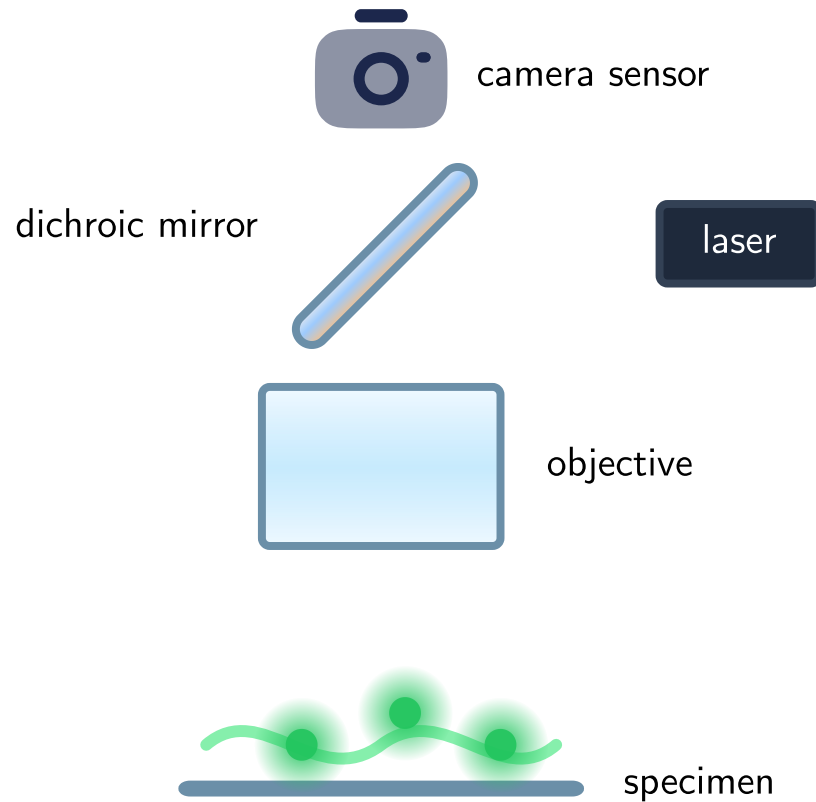
Can we circumvent this limit ?



Optical microscope

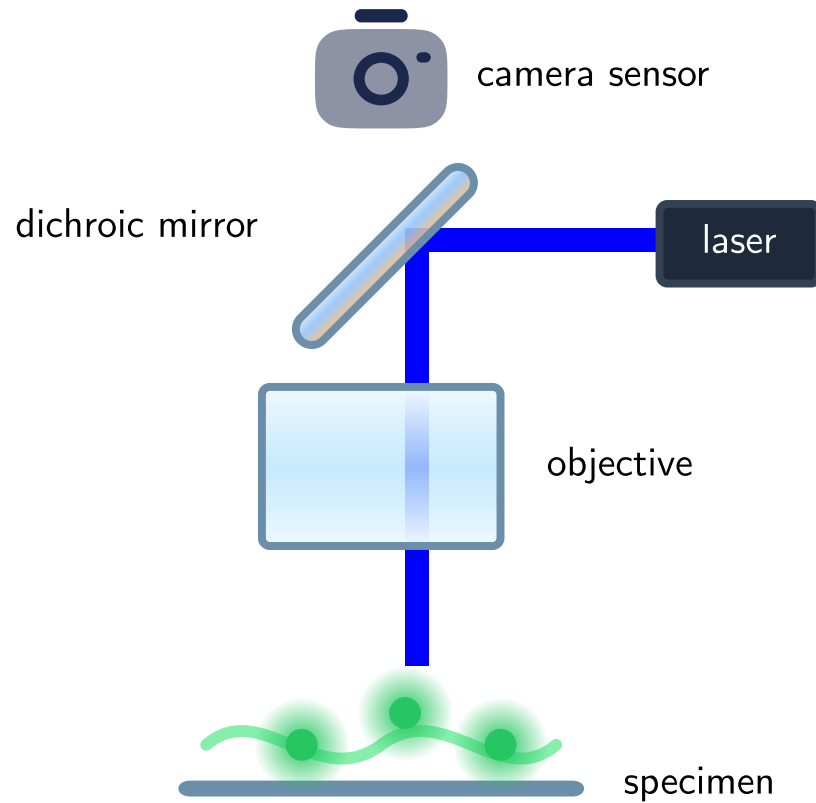
Fluorescence microscopy

Fluorescence microscopy exploits fluorophores stained on the specimen to super-resolve acquisitions.



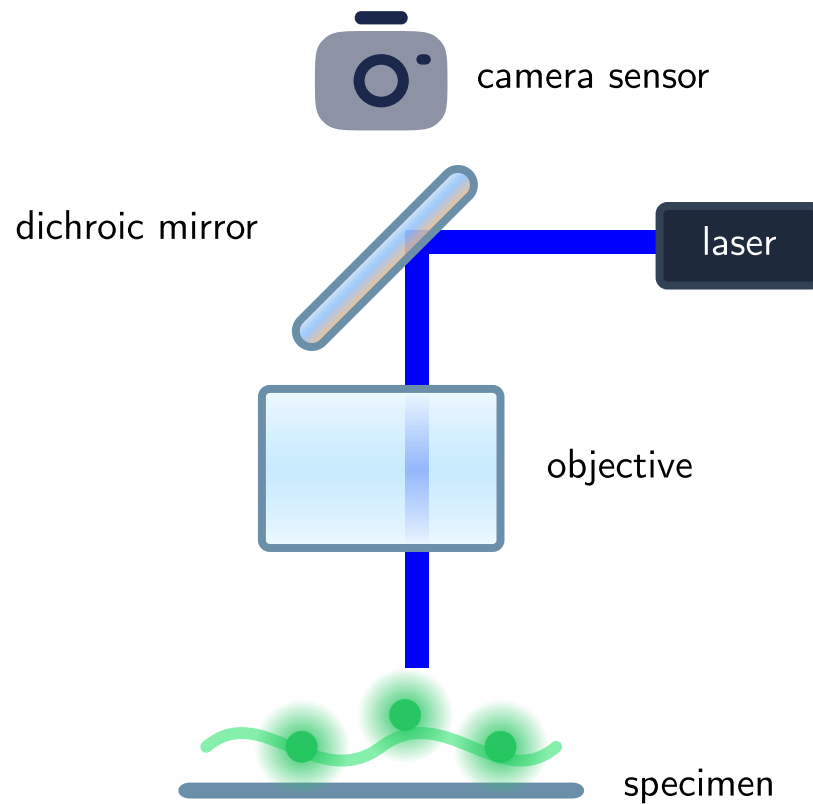
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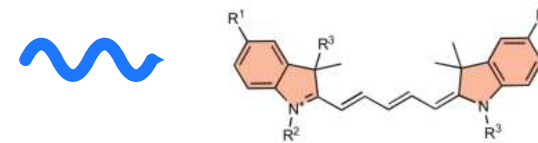


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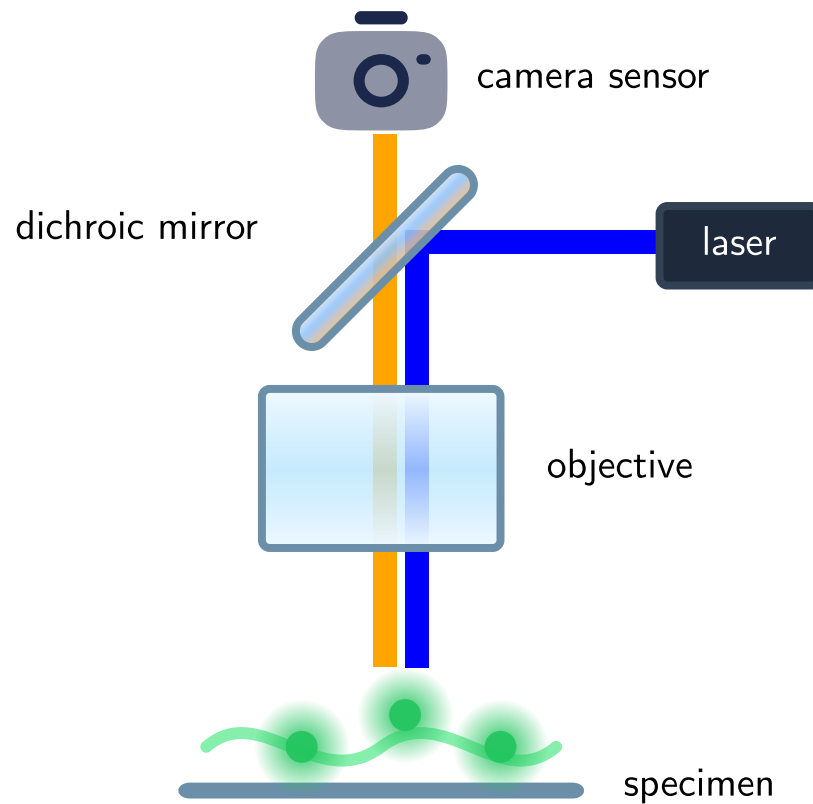


Fluorophores are fluorescent chemical compounds that once excited stochastically re-emit light at a longer wavelength.

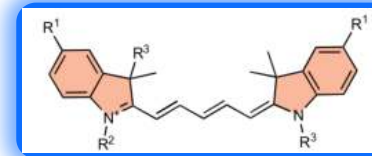


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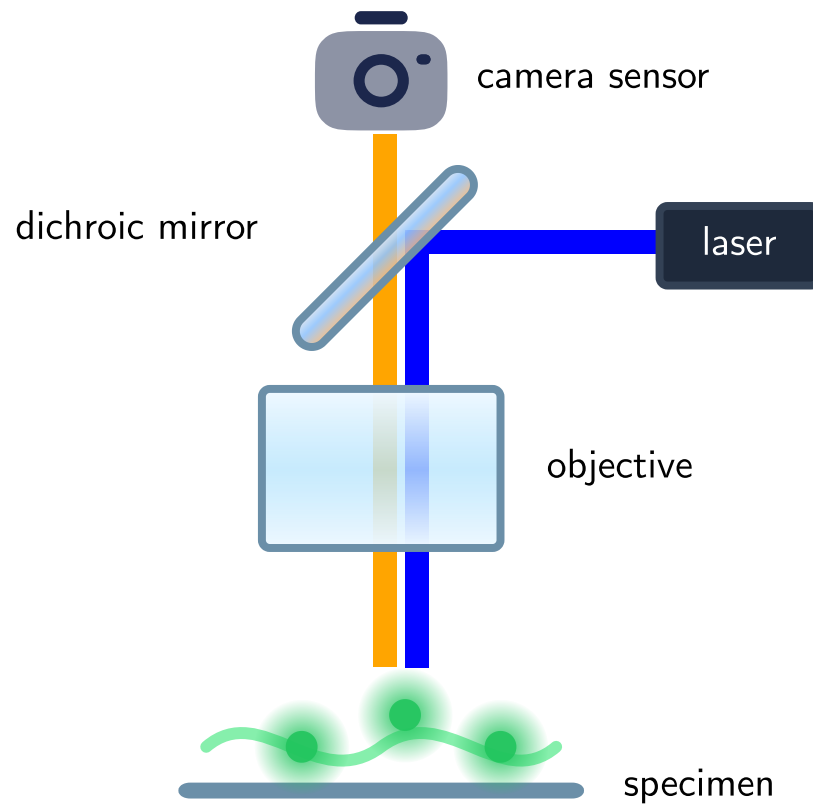


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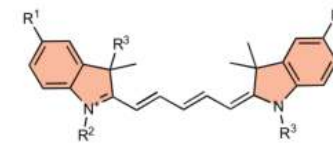


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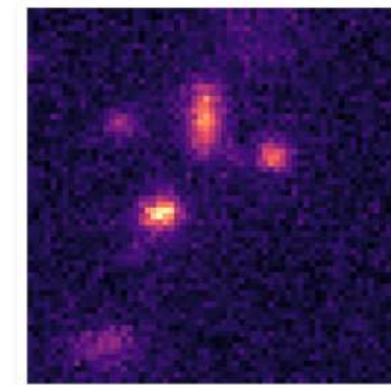
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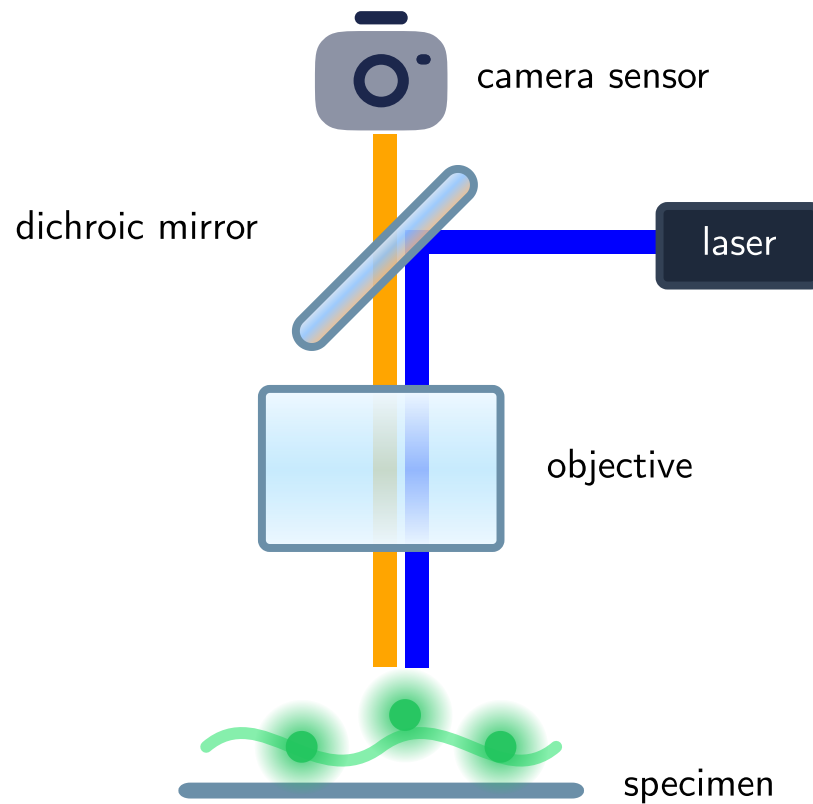


We observe a **sequence of images**, showing stochastically blinking "blobs".

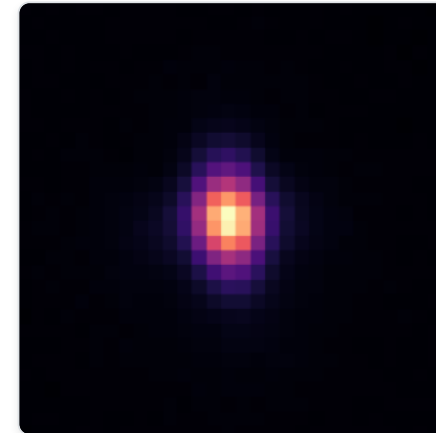


3D Point spread function

Fluorescence microscopy exploits fluorophores stained on the specimen to super-resolve acquisitions.

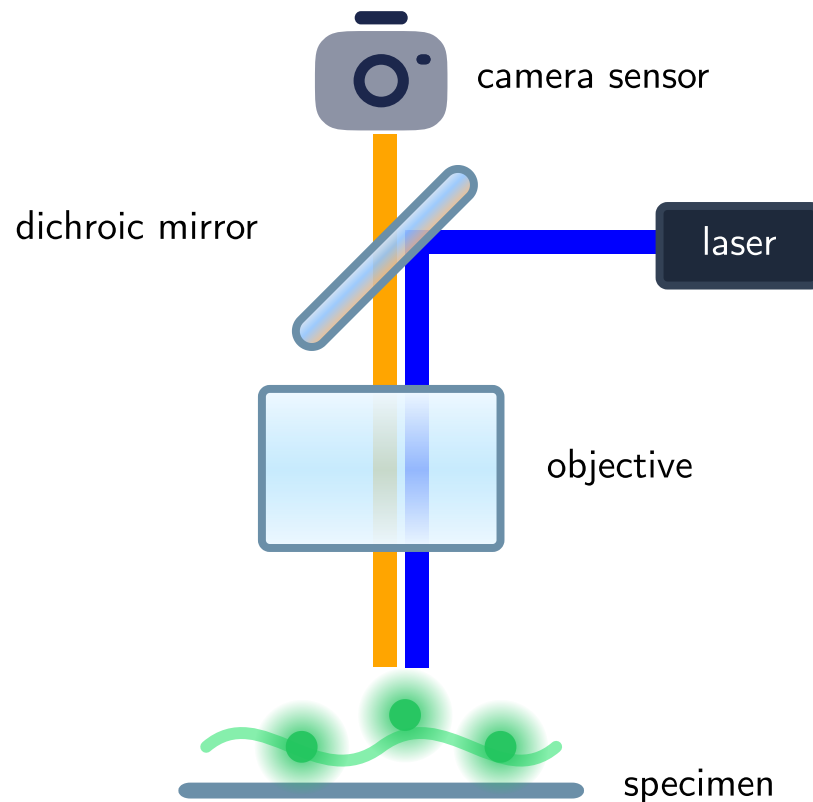


Each "blob" is the **point spread function (PSF)**: the image of a single point source formed by the microscope.

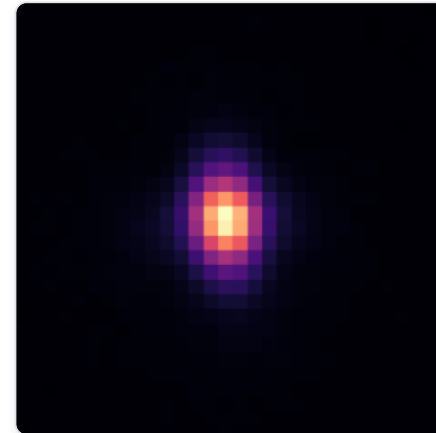


3D Point spread function

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We can use **cylindrical lenses** in the objective to introduce **astigmatism**, which breaks the axial symmetry.



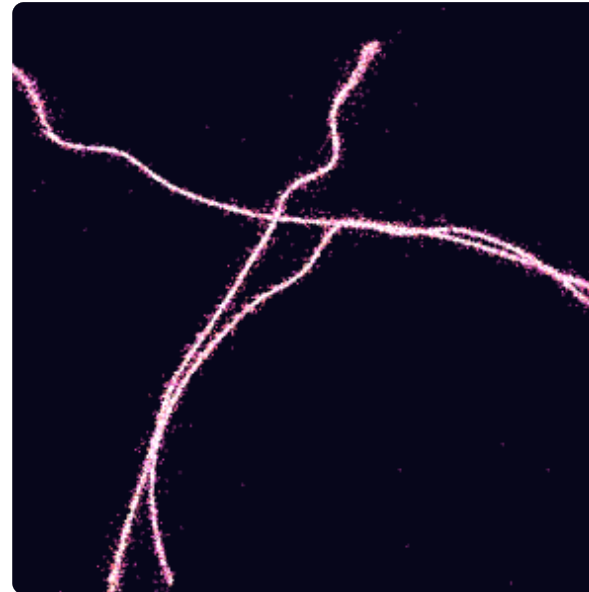
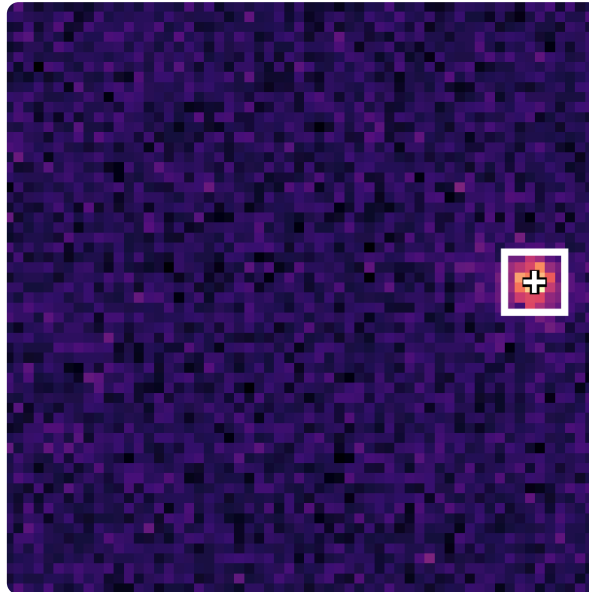
At the focal plane

Single molecule localization microscopy

Single molecule localization microscopy (SMLM) **detects** and **localizes** individual fluorophores to reconstruct a high-resolution image.



Frames ≤ 1

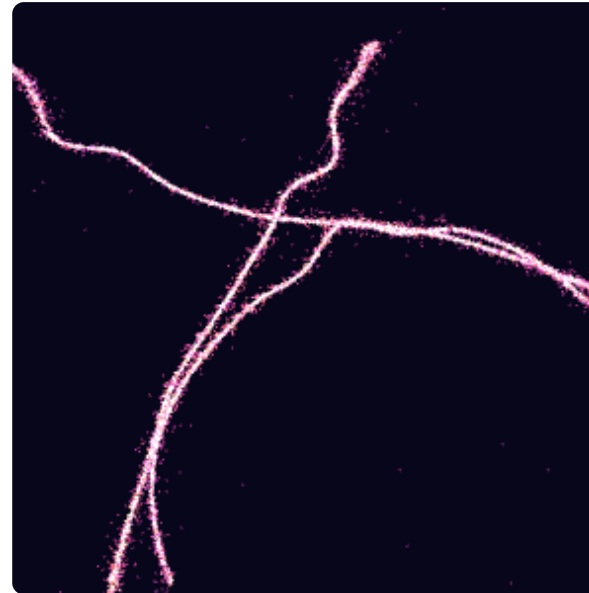
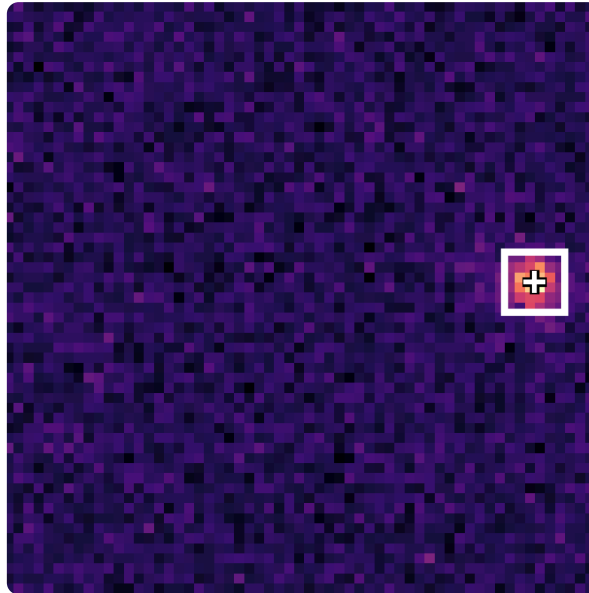


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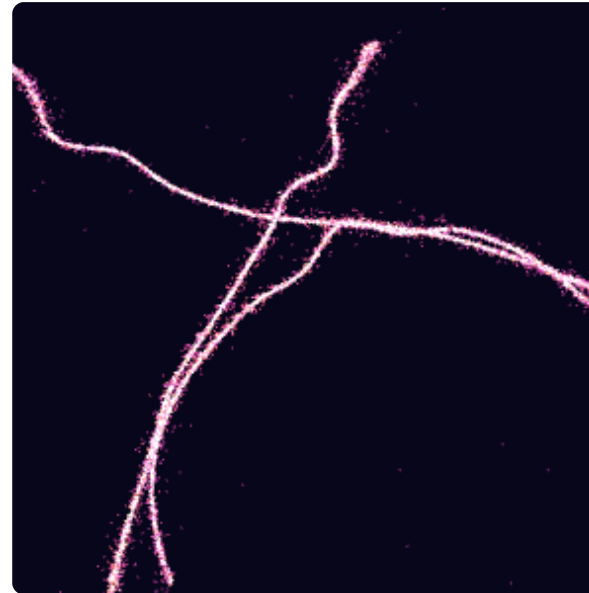
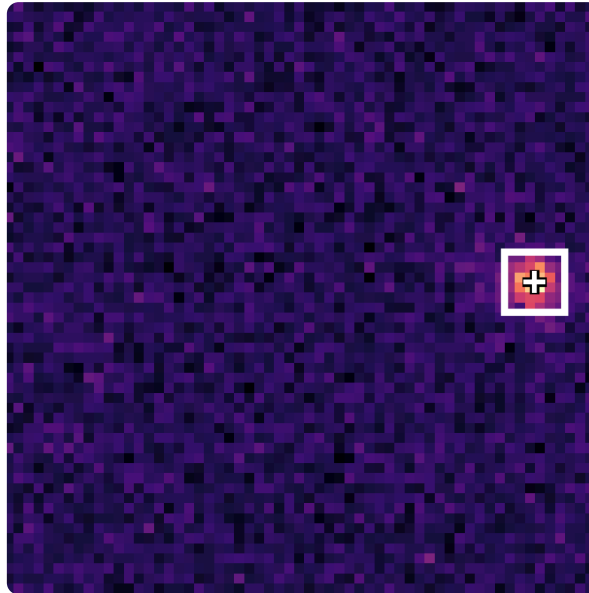
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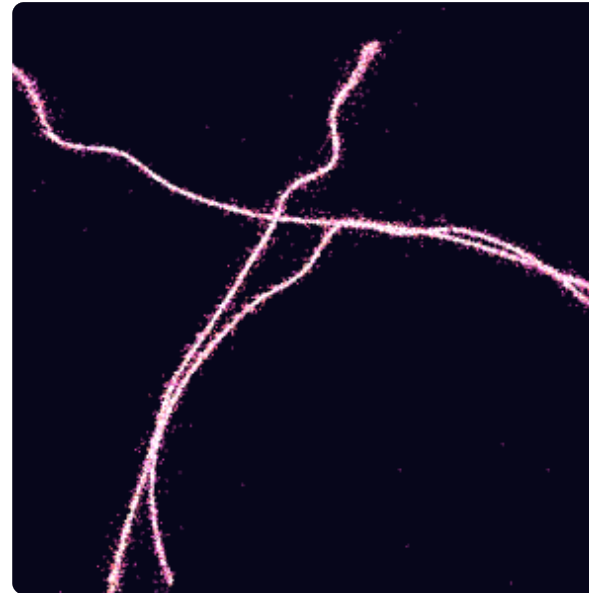
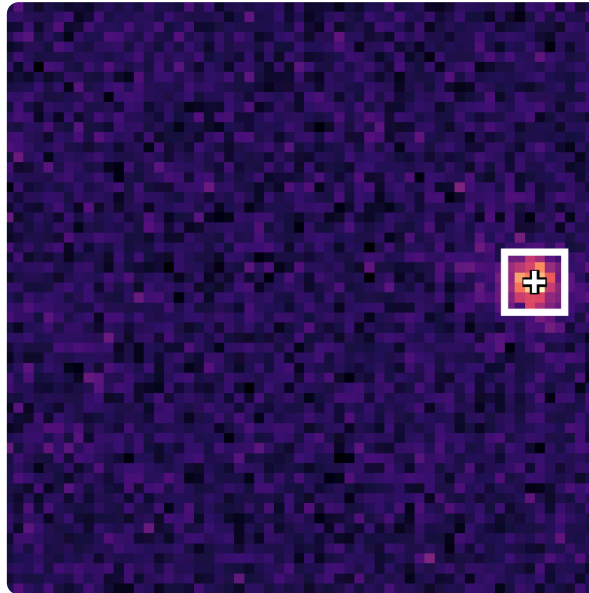
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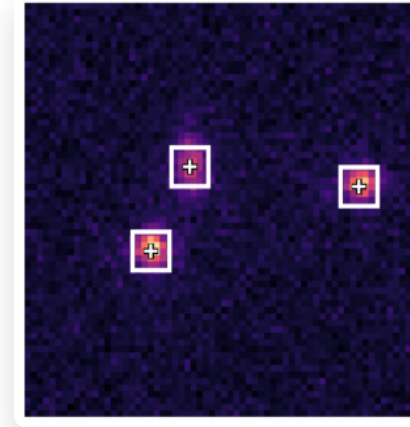
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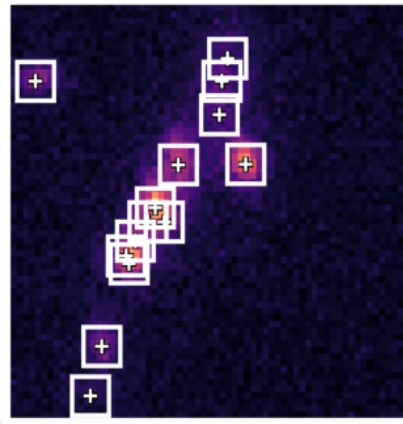
Slow acquisition 🐌

SMLM Density

Low Density: Laser power is tuned to limit the number of active fluorophores, such that **no two activations occurred simultaneously in the same diffraction-limited area** with high probability.



$0.2 \text{ act}/\mu\text{m}^2/\text{frame}$



$2.0 \text{ act}/\mu\text{m}^2/\text{frame}$

High Density: Overlapping activations create uncertainties in the number of fluorophores, leading to **localization errors** and a degraded reconstruction.

Single molecule localisation microscopy

+ Advantages

- **Excellent resolution**, $d \simeq 30\text{nm}$ in a classic setup.
Can reach 10nm if the density is low enough.
- **Less phototoxicity** for photosensitive specimens.
(compared to active methods like STED)

- Inconvenients

- Limited to "low" activation densities
Otherwise, precise individual localization becomes impossible.
- Requires thousands of frames
Imaging process can take minutes to hours.
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Can learning-based methods enable faster SMLM reconstruction ?

Optimal transport unlocks end-to-end learning for single-molecule localization

Romain Séailles, Jean-Baptiste Masson, Jean Ponce, and Julien Mairal.

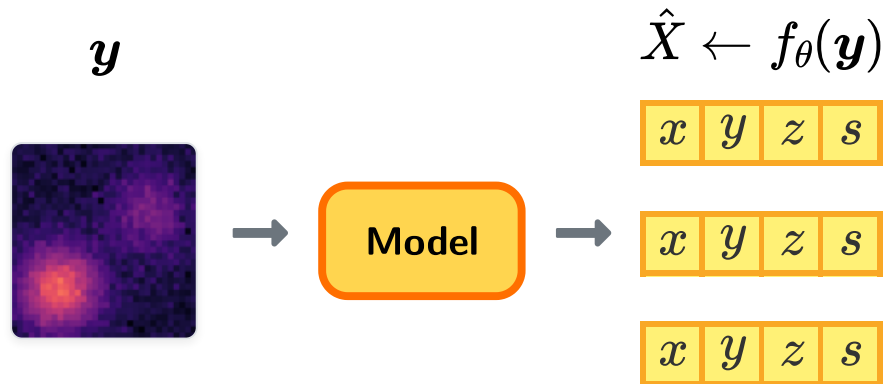
2026

to appear in ICLR

Joint detection and localization framework

Given a frame $\mathbf{y}$, a model f_θ predicts a set of candidates $\hat{X} = \{(x_i, y_i, z_i, s_i)\}_{1 \leq i \leq d}$ where:

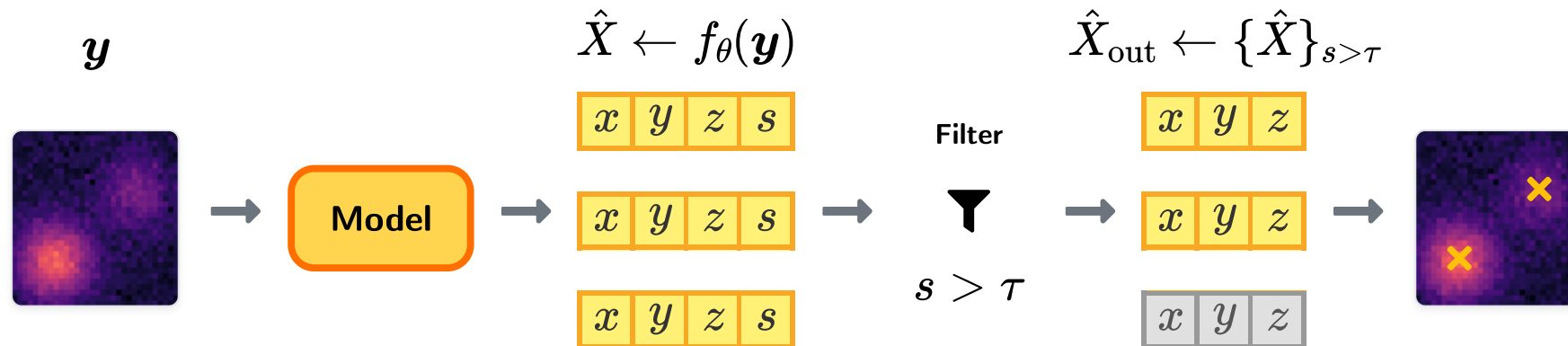
 (x_i, y_i, z_i) : 3D spatial coordinates  $s_i \in [0, 1]$: Detection confidence score



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 At inference, candidates are filtered by score ($s > \tau$)
→ this threshold controls the Precision-Recall trade-off.

Supervised learning

Following [Speiser et al. \(2021\)](#), we propose to train our model using **supervised learning**.

Mathematically, we **minimize the empirical risk** via **stochastic gradient descent**:

$$\theta^* = \operatorname{argmin}_{\theta \in \Theta} \mathbb{E}_{X, \mathbf{y} \sim \mathcal{D}} [\mathcal{L}(f_{\theta}(\mathbf{y}), X)]$$

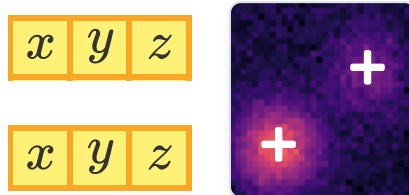
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$$X, y \sim \mathcal{D}$$



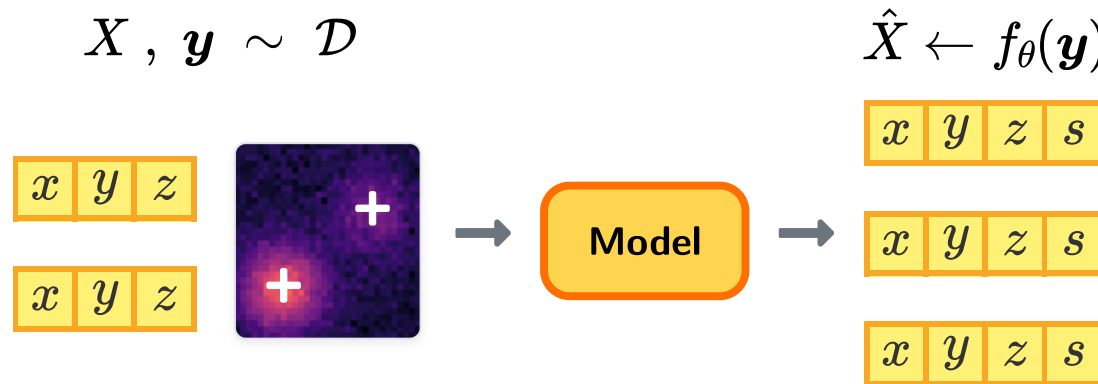
 A fluorophores-image pair dataset

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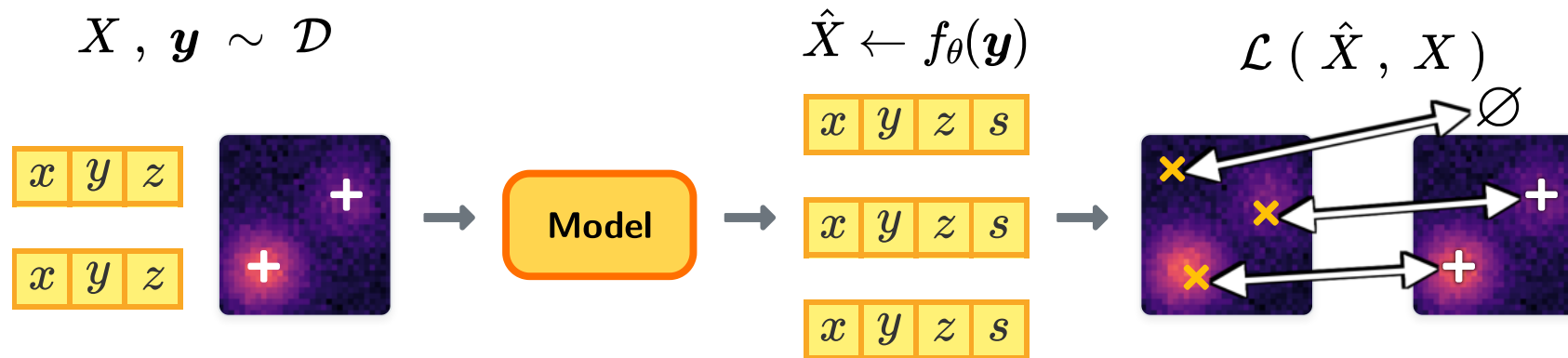
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 A loss function

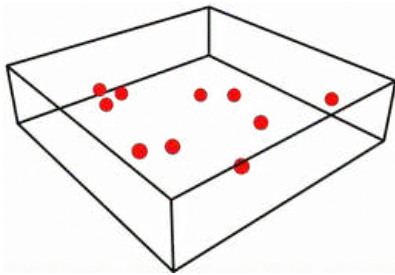
Synthetic data

Ground truth unavailable $\Rightarrow$ simulator built on the model of [Sage et al. \(2019\)](#).

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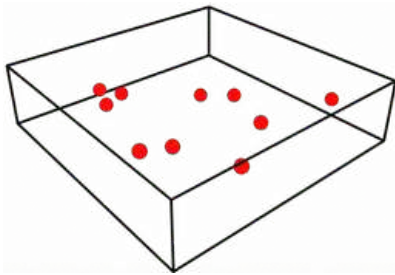


Uniform 3D spatial distribution.

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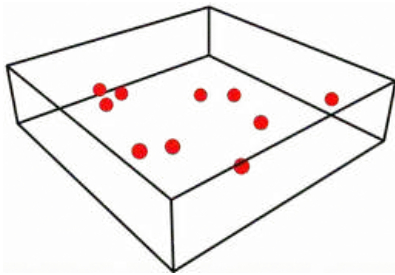


Markovian dynamics with exponential lifetimes.

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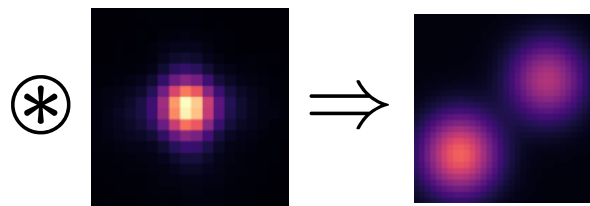
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3. Apply PSF

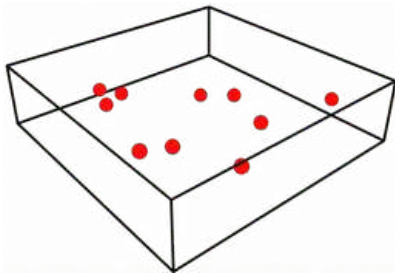


Empirical PSF calibrated on real data.

Synthetic data

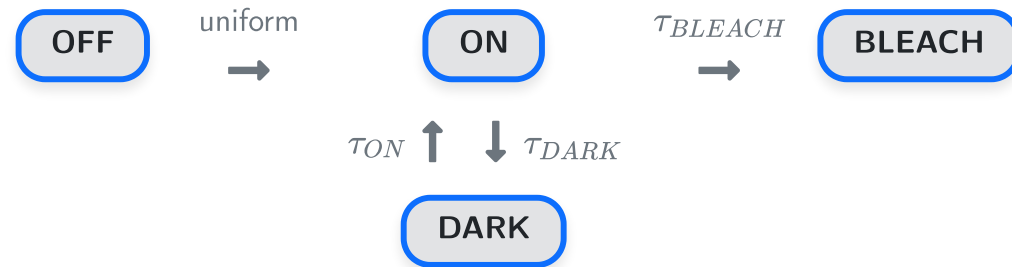
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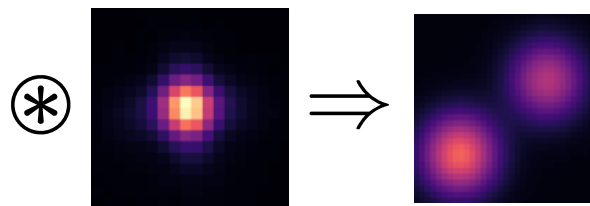
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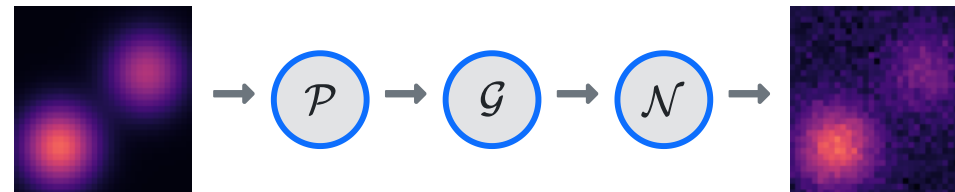
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Empirical PSF calibrated on real data.

4. Camera model



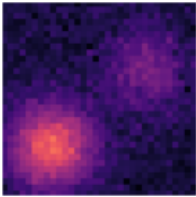
Composite noise includes shot noise (Poisson dist.), amplification noise (Gamma dist.) and readout noise (normal dist.).

Model architecture

Inspired by the success of iterative refinement networks for inverse problems [[1](#),[2](#),[3](#)], we proposed an iterative architecture that leverages the image formation process.

Model architecture

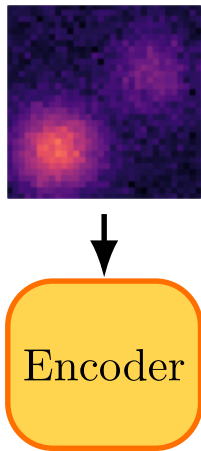
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Input image $y \in \mathbb{R}^{H \times W}$.

Model architecture

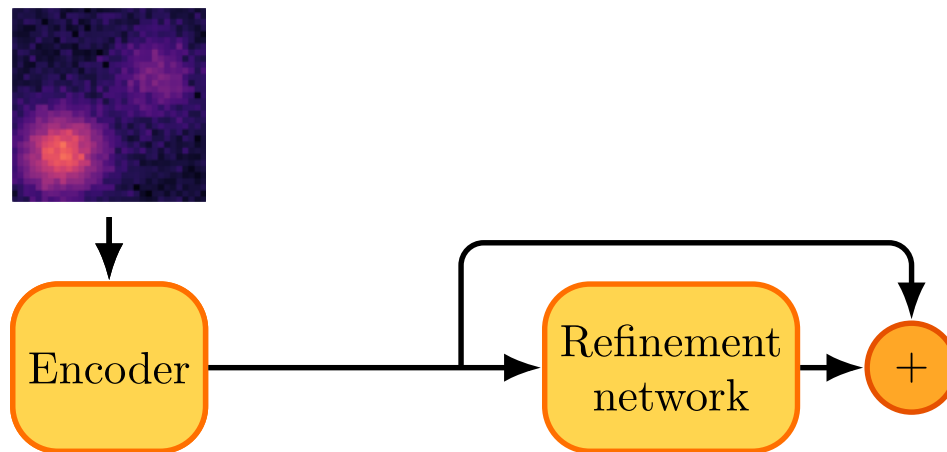
Inspired by the success of iterative refinement networks for inverse problems [1,2,3], we proposed an iterative architecture that leverages the image formation process.



The encoder is implemented as a 2-layer U-Net, mapping $\mathbf{y}$ to a latent representation $\mathbf{z} \in \mathbb{R}^{C \times H \times W}$.

Model architecture

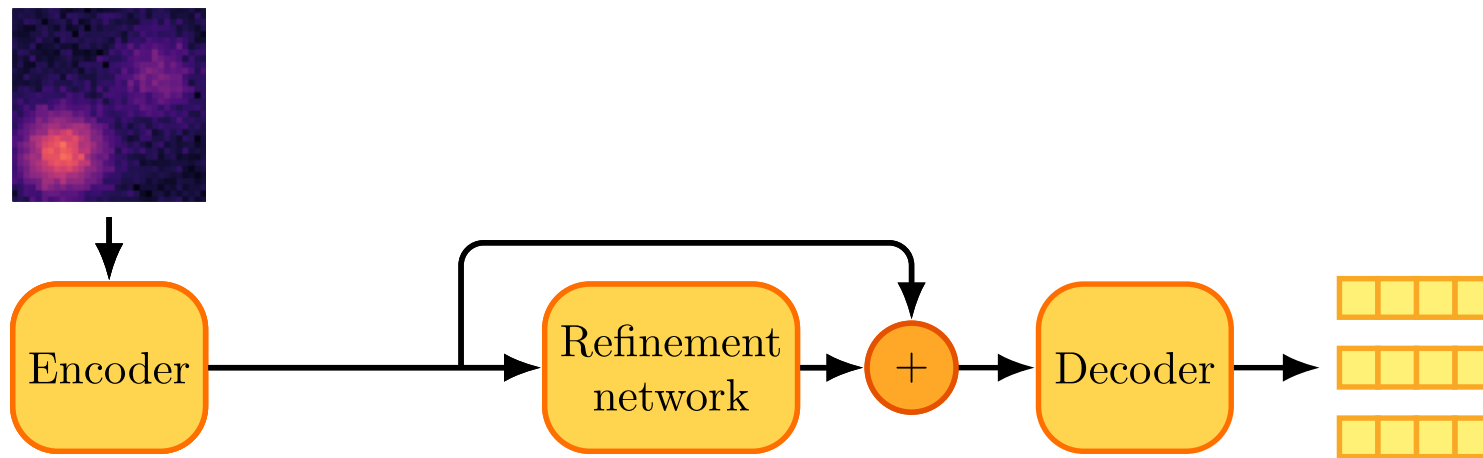
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The core network is also implemented as a 2-layer U-Net.

Model architecture

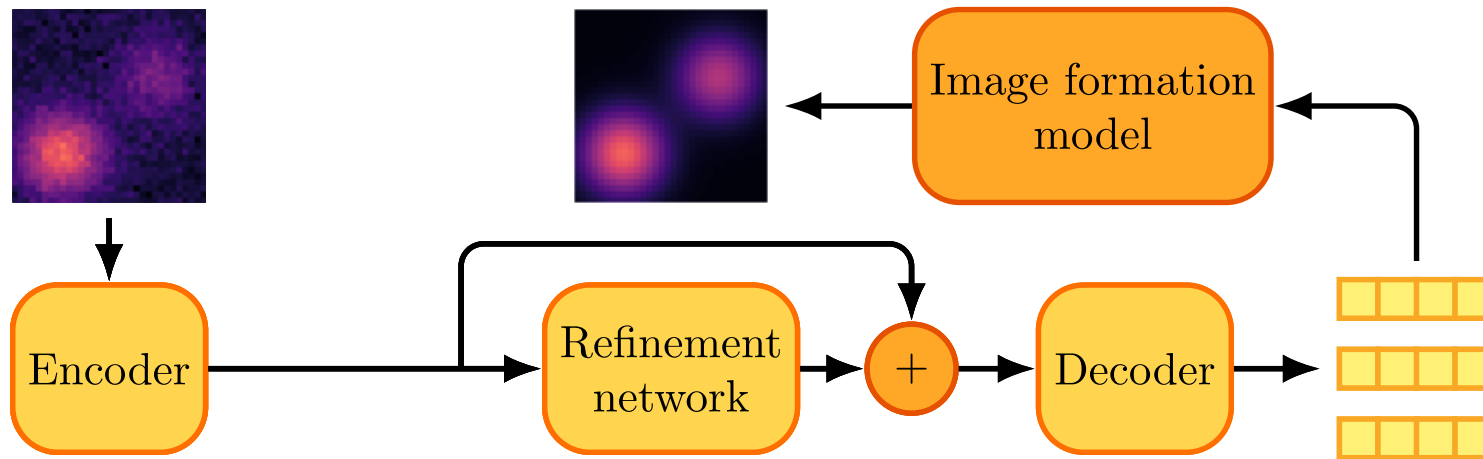
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The decoder is implemented as a shallow CNN with max-pooling followed by flattening, yielding $\hat{X} \in \mathbb{R}^{d \times 4}$.

Model architecture

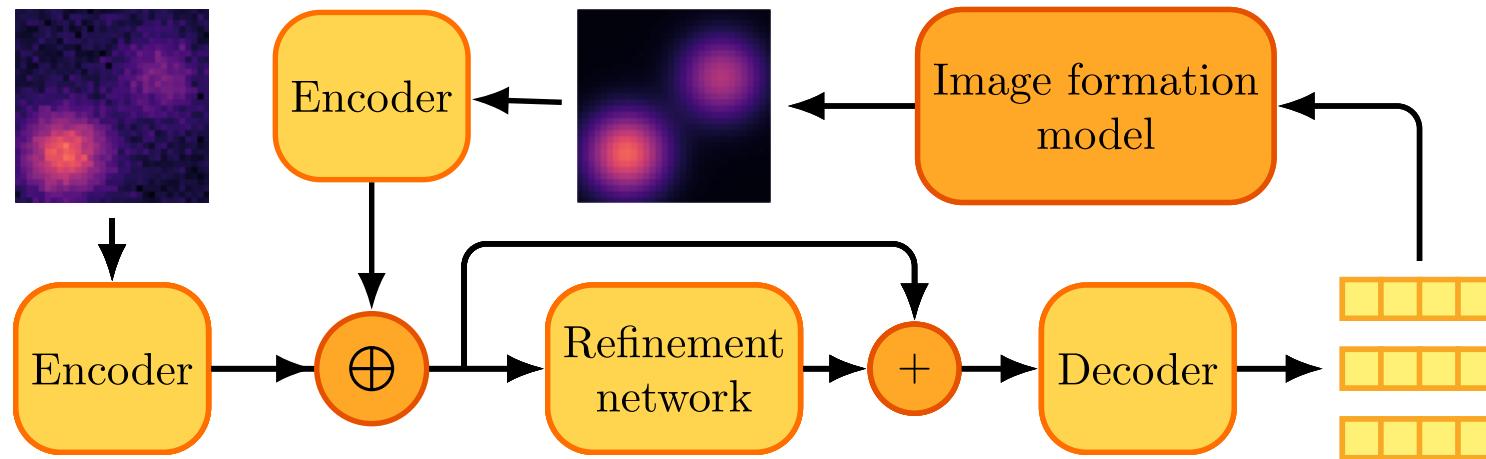
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We render the output $\hat{X}$ into a new image $\hat{y}$ with the image formation process described earlier (using relaxation tricks on detection scores to preserve differentiability).

Model architecture

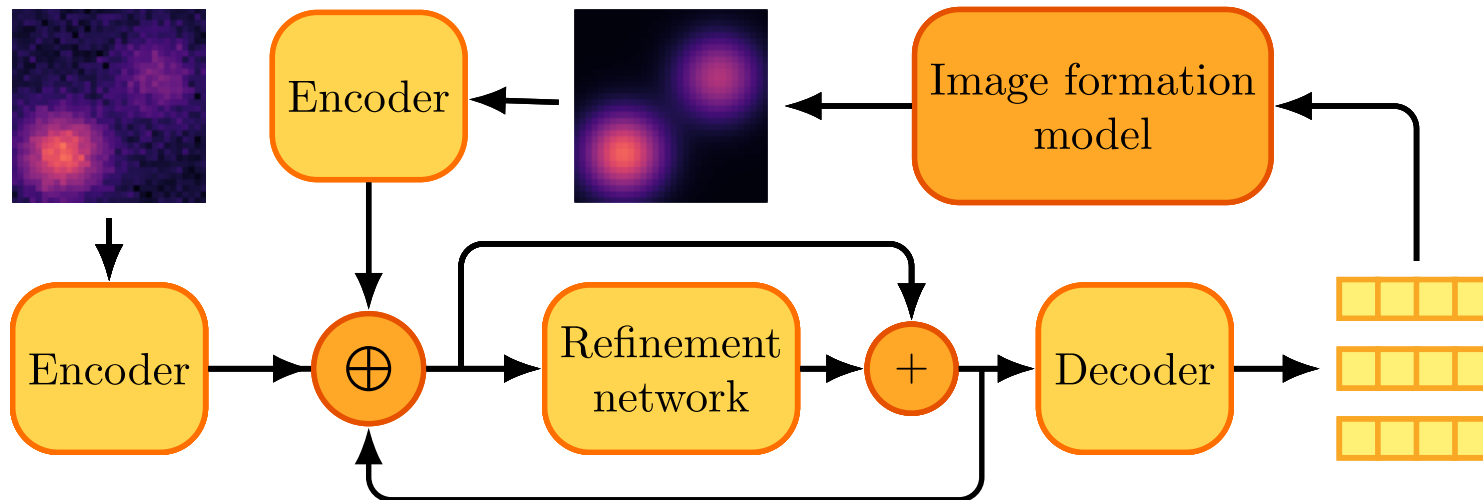
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The same encoder maps the rendered image $\hat{y}$ to a latent representation $\hat{z} \in \mathbb{R}^{C \times H \times W}$. The original and new latents, z and $\hat{z}$, are concatenated for refinement in the next iteration.

Model architecture

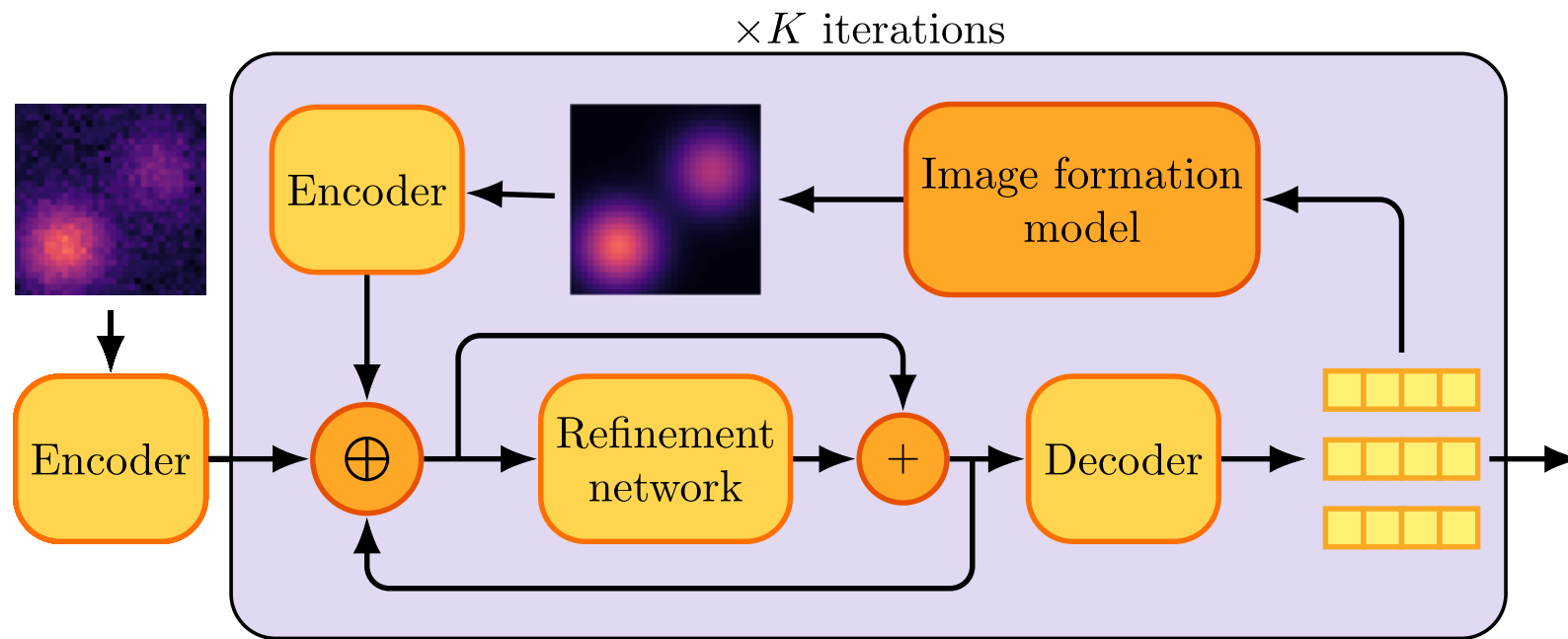
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The refinement pass, alongside the original and rendered latent representation, also needs the current latent state to effectively refine it.

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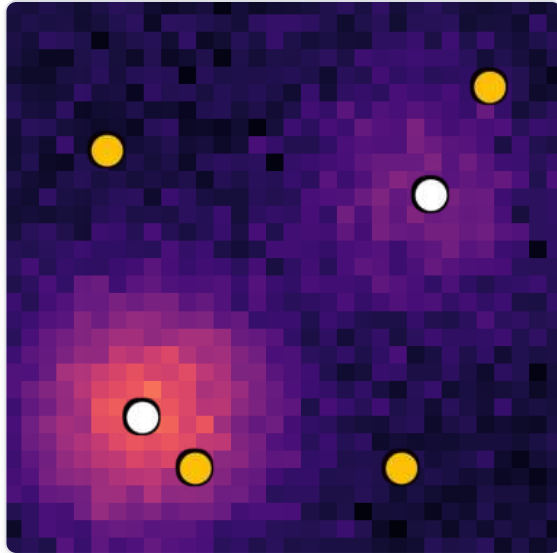


This iterative refinement scheme runs for $K = 3$ iterations in practice.

Loss function

We have constructed a loss function using **regularized-bipartite matching**.

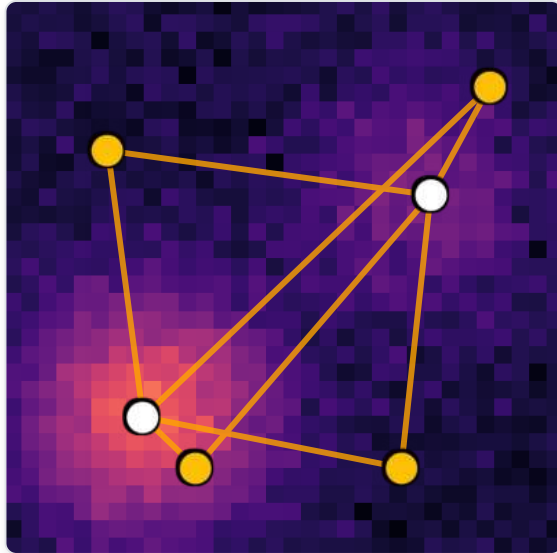
● Prediction ○ Ground truth



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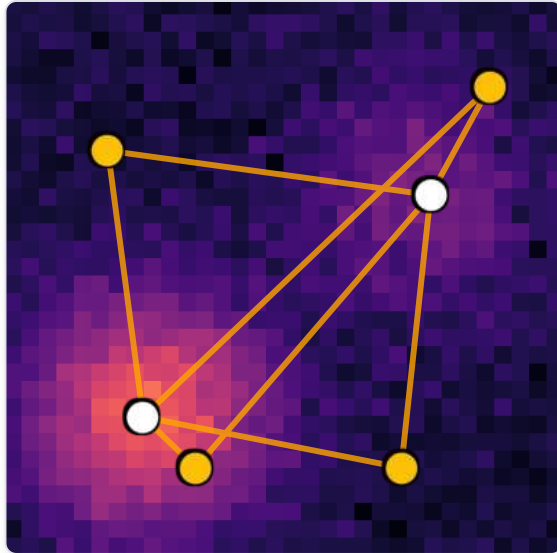
- Compute pairwise localization costs (with negative normal log-likelihood):

$$l_{i,j} = (\hat{\mathbf{x}}_i - \mathbf{x}_j)^\top \Sigma^{-1} (\hat{\mathbf{x}}_i - \mathbf{x}_j) + \log \det(\Sigma)$$

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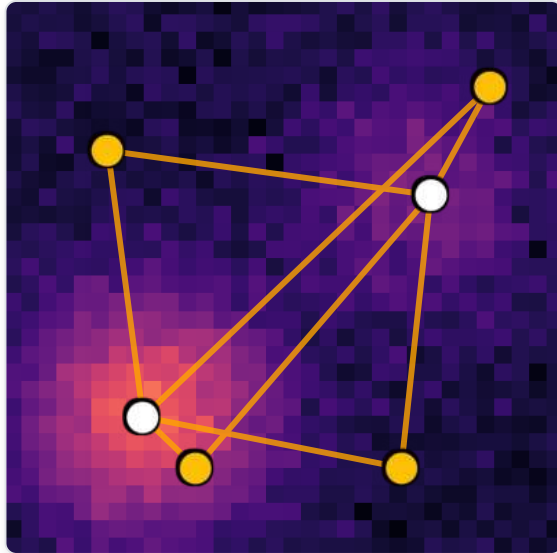
- Create a squared cost matrix with localization cost and detection penalty (with binary cross-entropy):

$$\mathbf{C} = \begin{bmatrix} l_{1,1} - \log(\hat{s}_1) & l_{2,1} - \log(\hat{s}_2) & l_{3,1} - \log(\hat{s}_3) & l_{4,1} - \log(\hat{s}_4) \\ l_{1,2} - \log(\hat{s}_1) & l_{2,2} - \log(\hat{s}_2) & l_{3,2} - \log(\hat{s}_3) & l_{4,2} - \log(\hat{s}_4) \\ -\log(1 - \hat{s}_1) & -\log(1 - \hat{s}_2) & -\log(1 - \hat{s}_3) & -\log(1 - \hat{s}_4) \\ -\log(1 - \hat{s}_1) & -\log(1 - \hat{s}_2) & -\log(1 - \hat{s}_3) & -\log(1 - \hat{s}_4) \end{bmatrix}$$

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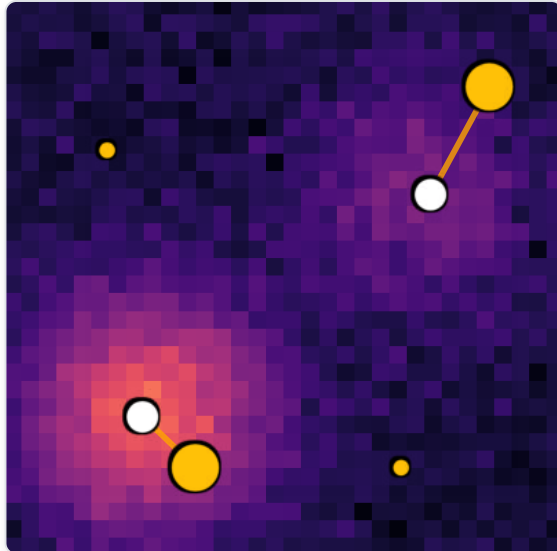
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$$\mathcal{L}(\hat{X}, X) = \langle \Gamma^* \mid C \rangle_{\mathcal{F}}, \quad \text{where } \Gamma^* = \arg \min_{\Gamma \in \mathbb{B}} \langle \Gamma \mid C \rangle_{\mathcal{F}} - \epsilon H(\Gamma),$$

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We have constructed a loss function using **regularized-bipartite matching**.

● Prediction ○ Ground truth



- Compute pairwise localization costs (with negative normal log-likelihood):

$$l_{i,j} = (\hat{\mathbf{x}}_i - \mathbf{x}_j)^\top \Sigma^{-1} (\hat{\mathbf{x}}_i - \mathbf{x}_j) + \log \det(\Sigma)$$

- Create a squared cost matrix with localization cost and detection penalty (with binary cross-entropy):

$$\mathbf{C} = \begin{bmatrix} l_{1,1} - \log(\hat{s}_1) & l_{2,1} - \log(\hat{s}_2) & \mathbf{l_{3,1} - \log(\hat{s}_3)} & l_{4,1} - \log(\hat{s}_4) \\ \mathbf{l_{1,2} - \log(\hat{s}_1)} & l_{2,2} - \log(\hat{s}_2) & l_{3,2} - \log(\hat{s}_3) & l_{4,2} - \log(\hat{s}_4) \\ -\log(1 - \hat{s}_1) & \mathbf{-\log(1 - \hat{s}_2)} & -\log(1 - \hat{s}_3) & -\log(1 - \hat{s}_4) \\ -\log(1 - \hat{s}_1) & -\log(1 - \hat{s}_2) & -\log(1 - \hat{s}_3) & \mathbf{-\log(1 - \hat{s}_4)} \end{bmatrix}$$

$$\mathcal{L}(\hat{X}, X) = \langle \Gamma^* \mid C \rangle_{\mathcal{F}}, \quad \text{where } \Gamma^* = \arg \min_{\Gamma \in \mathbb{B}} \langle \Gamma \mid C \rangle_{\mathcal{F}} - \epsilon H(\Gamma),$$

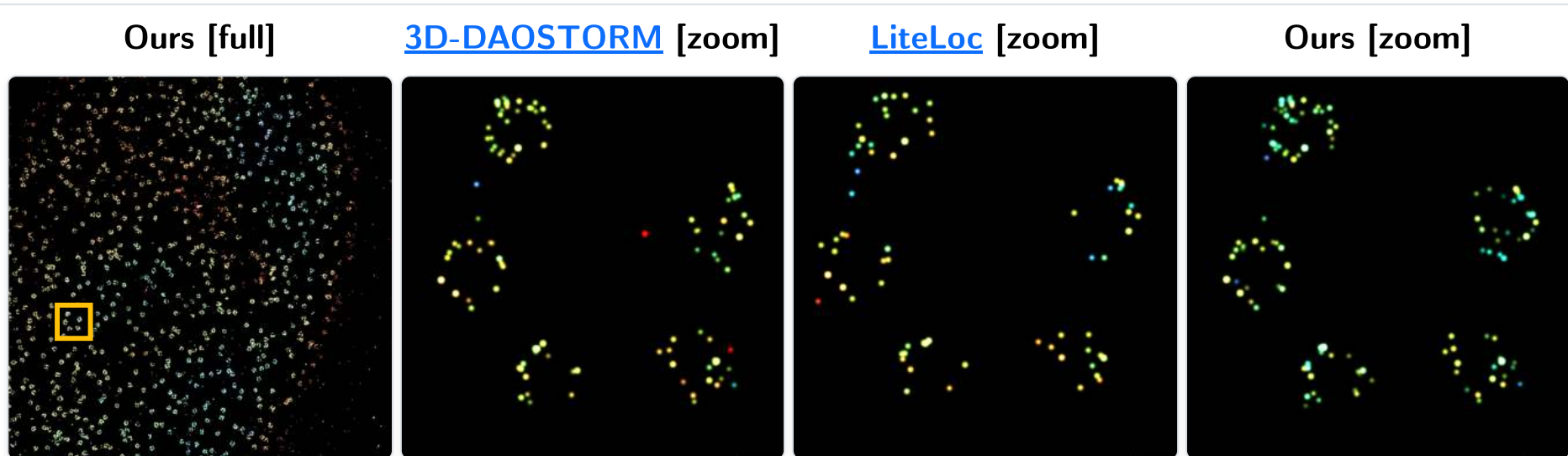
Synthetic results

Density	SNR	Method	Precision $\uparrow$	Recall $\uparrow$	Jaccard $\uparrow$	RMSE _{lat} $\downarrow$	RMSE _{ax} $\downarrow$	E _{3D} $\uparrow$
0.2	High	3D-DAOSTORM	0.964	0.919	0.914	11.9	16.9	0.821
		DECODE	0.961 $\pm$ 0.003	0.998 $\pm$ 0.001	0.959 $\pm$ 0.003	8.8 $\pm$ 0.1	10.7 $\pm$ 0.1	0.895 $\pm$ 0.003
		LiteLoc	0.996 $\pm$ 0.002	0.987 $\pm$ 0.001	0.983 $\pm$ 0.001	9.0 $\pm$ 0.1	11.7 $\pm$ 0.1	0.912 $\pm$ 0.001
		Ours	0.998 $\pm$ 0.002	0.978 $\pm$ 0.016	0.980 $\pm$ 0.010	7.5 $\pm$ 0.4	10.0 $\pm$ 3.5	0.920 $\pm$ 0.007
	Low	3D-DAOSTORM	0.978	0.835	0.833	19.3	29.8	0.685
		DECODE	0.918 $\pm$ 0.002	0.978 $\pm$ 0.001	0.903 $\pm$ 0.002	20.5 $\pm$ 0.1	26.2 $\pm$ 0.1	0.757 $\pm$ 0.001
		LiteLoc	0.995 $\pm$ 0.001	0.939 $\pm$ 0.001	0.934 $\pm$ 0.001	17.0 $\pm$ 0.1	25.0 $\pm$ 0.4	0.798 $\pm$ 0.001
		Ours	0.985 $\pm$ 0.001	0.961 $\pm$ 0.001	0.947 $\pm$ 0.001	18.8 $\pm$ 0.2	24.5 $\pm$ 0.1	0.802 $\pm$ 0.002
2.0	High	3D-DAOSTORM	0.914	0.678	0.643	56.8	76.6	0.373
		DECODE	0.923 $\pm$ 0.003	0.946 $\pm$ 0.002	0.876 $\pm$ 0.004	32.2 $\pm$ 0.3	33.0 $\pm$ 0.4	0.706 $\pm$ 0.004
		LiteLoc	0.993 $\pm$ 0.001	0.863 $\pm$ 0.002	0.858 $\pm$ 0.001	30.7 $\pm$ 0.2	36.0 $\pm$ 0.3	0.699 $\pm$ 0.001
		Ours	0.992 $\pm$ 0.002	0.895 $\pm$ 0.011	0.883 $\pm$ 0.007	24.8 $\pm$ 0.6	28.4 $\pm$ 0.5	0.750 $\pm$ 0.004
	Low	3D-DAOSTORM	0.914	0.496	0.475	74.4	120.0	0.116
		DECODE	0.859 $\pm$ 0.034	0.874 $\pm$ 0.006	0.756 $\pm$ 0.027	56.4 $\pm$ 0.3	65.3 $\pm$ 0.4	0.468 $\pm$ 0.008
		LiteLoc	0.992 $\pm$ 0.001	0.729 $\pm$ 0.002	0.725 $\pm$ 0.001	46.2 $\pm$ 0.4	63.8 $\pm$ 0.2	0.500 $\pm$ 0.002
		Ours	0.973 $\pm$ 0.003	0.812 $\pm$ 0.007	0.794 $\pm$ 0.005	48.4 $\pm$ 0.6	59.5 $\pm$ 0.4	0.536 $\pm$ 0.003
8.0	High	3D-DAOSTORM	0.910	0.392	0.379	83.3	133.9	0.009
		DECODE	0.973 $\pm$ 0.001	0.627 $\pm$ 0.002	0.617 $\pm$ 0.002	59.58 $\pm$ 0.02	71.5 $\pm$ 0.5	0.371 $\pm$ 0.006
		LiteLoc	0.988 $\pm$ 0.001	0.557 $\pm$ 0.003	0.553 $\pm$ 0.003	60.4 $\pm$ 0.2	80.5 $\pm$ 0.5	0.319 $\pm$ 0.004
		Ours	0.989 $\pm$ 0.001	0.578 $\pm$ 0.008	0.574 $\pm$ 0.008	52.5 $\pm$ 0.4	64.3 $\pm$ 0.7	0.384 $\pm$ 0.003
	Low	3D-DAOSTORM	0.908	0.211	0.217	92.6	175.8	-0.216
		DECODE	0.93 $\pm$ 0.03	0.415 $\pm$ 0.005	0.402 $\pm$ 0.009	80.25 $\pm$ 0.04	105.2 $\pm$ 1.0	0.090 $\pm$ 0.007
		LiteLoc	0.986 $\pm$ 0.001	0.339 $\pm$ 0.002	0.338 $\pm$ 0.002	76.1 $\pm$ 0.1	110.1 $\pm$ 0.8	0.055 $\pm$ 0.002
		Ours	0.983 $\pm$ 0.002	0.376 $\pm$ 0.008	0.374 $\pm$ 0.008	74.3 $\pm$ 0.5	99.4 $\pm$ 1.1	0.103 $\pm$ 0.002

Long story short, **it works** on **synthetic benchmarks** at low and high densities.

Real results

We have compared reconstructions from different SMLM methods.

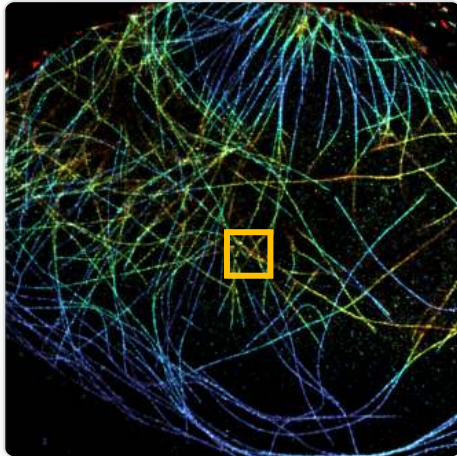


Our method yields a sharper reconstruction of NPC protein structures.

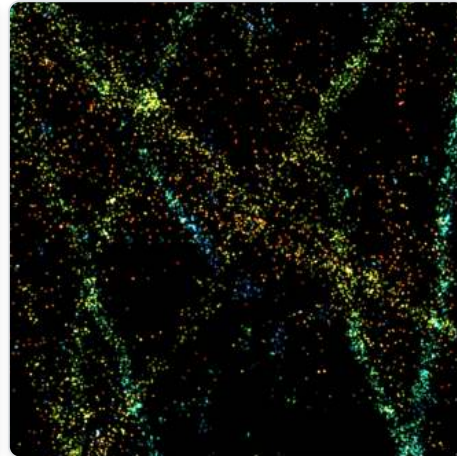
Real results

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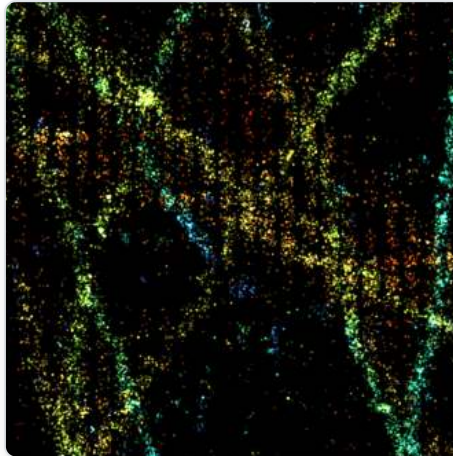
Ours [full]



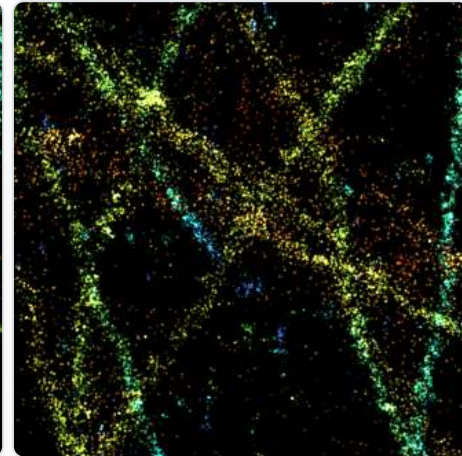
[3D-DAOSTORM](#) [zoom]



[LiteLoc](#) [zoom]



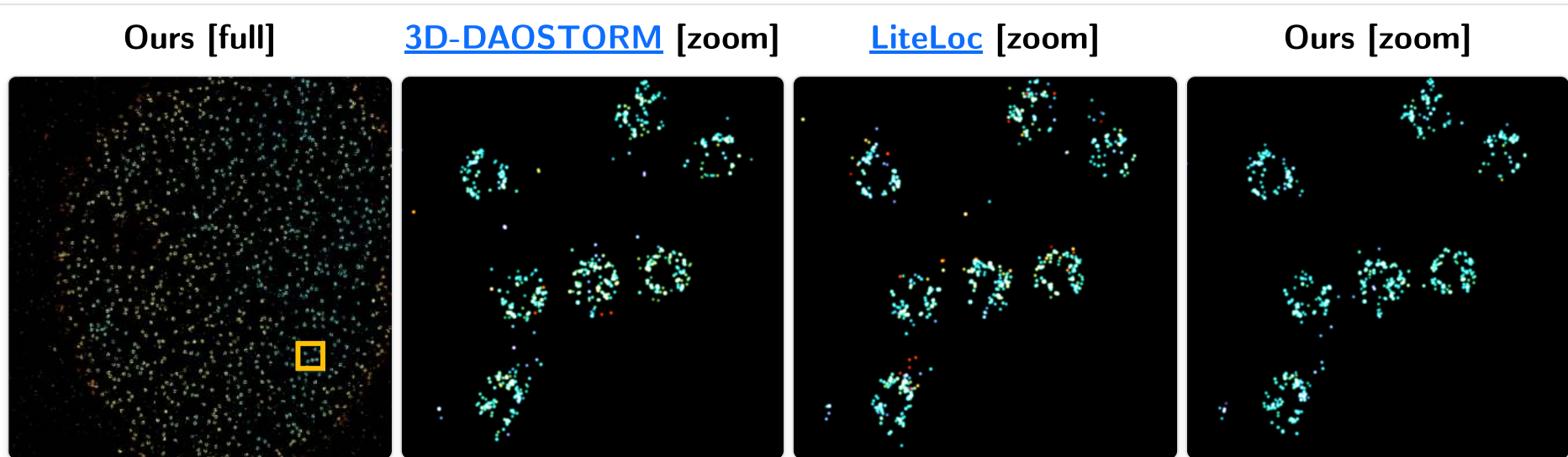
Ours [zoom]



It also proves efficient with microtubules, and avoids some artifacts present in other non-maximum suppression-based methods.

Real results

We have compared reconstructions from different SMLM methods.



In these reconstructions, depth is encoded via colors, and our method demonstrates better axial coherence than others.

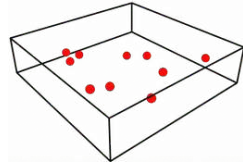
Limitations

Synthetic **training** and **inference** data **distributions** must be closely **matched**.

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3D-location (x, y, z)

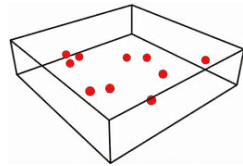


Is a uniform distribution the best option ?

Limitations

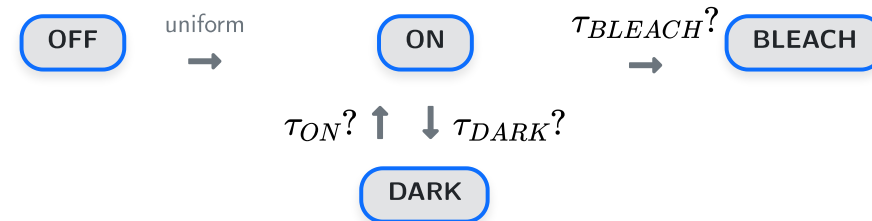
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Is a uniform distribution the best option ?

Temporal dynamics

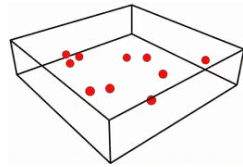


Are the time constants correct ?

Limitations

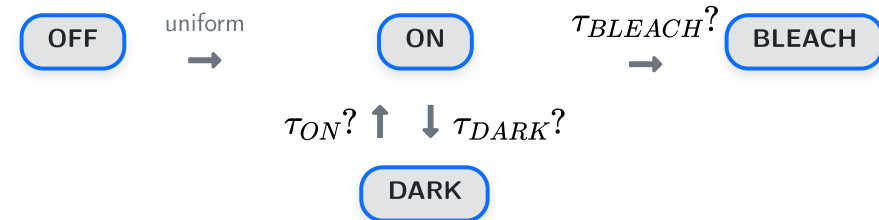
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3D-location (x, y, z)



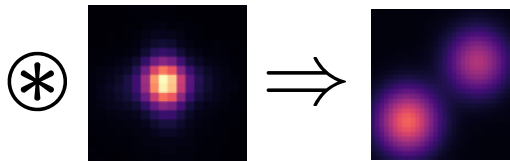
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Temporal dynamics



Are the time constants correct ?

PSF

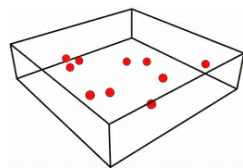


Is the PSF model accurate ?

Limitations

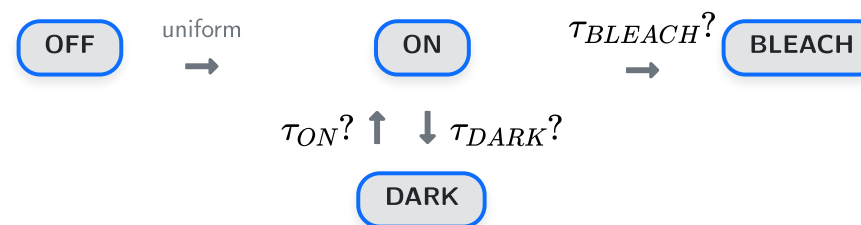
Synthetic **training** and **inference** data **distributions** must be closely **matched**.

3D-location (x, y, z)



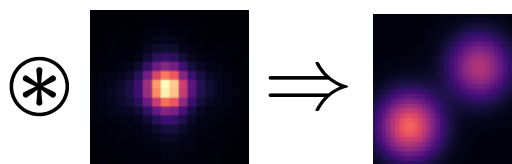
Is a uniform distribution the best option ?

Temporal dynamics



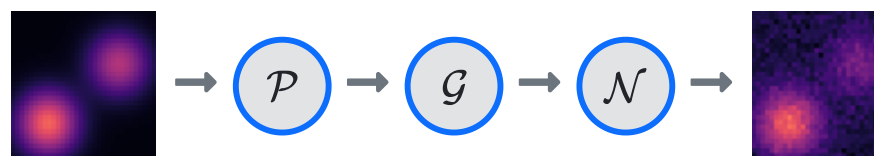
Are the time constants correct ?

PSF



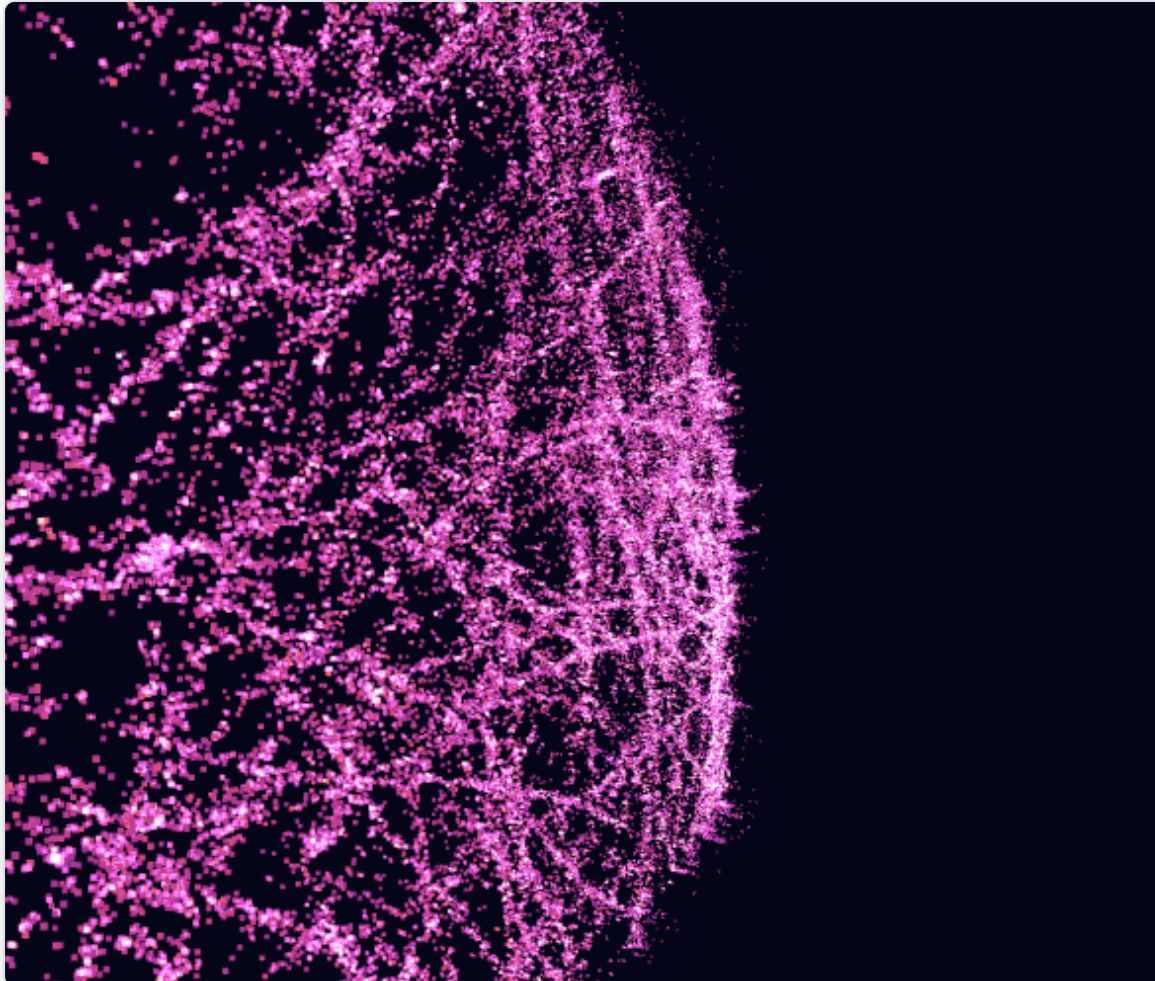
Is the PSF model accurate ?

Camera model



Is the camera model realistic ?

Grazie per l'attenzione



Any questions ?

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| PSL 

Inria